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Adaptive control of BTX removal in sulfur recovery units

Abstract

A system includes an outlet flowline from an activated carbon bed, a bypass flowline, a furnace, and a control system. The outlet flowline is configured to flow an acid gas stream out of the activated carbon bed after the activated carbon bed has removed benzene, toluene, and xylene from the acid gas stream. The bypass flowline includes a bypass valve and provides an alternative flow path for the acid gas stream exiting the activated carbon bed. The furnace includes a burner configured to combust at least a portion of the acid gas stream. The control system includes a temperature sensor and a controller. The temperature sensor is configured to measure a furnace temperature. The controller is configured to reduce a percent opening of the bypass valve based on the measured furnace temperature.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to managing benzene, toluene, and xylene (BTX) in acid gas.

BACKGROUND

(2) Hydrogen sulfide and carbon dioxide exist in various gas streams, including natural gas. Oil or gas that contains significant amounts of sulfur compounds like hydrogen sulfide is considered "sour", and oil refineries and gas processing plants utilize "sweetening" processes to remove such sulfur compounds. A typical sulfur recovery process includes the Claus process, which recovers elemental sulfur from gaseous hydrogen sulfide. The Claus process utilizes oxygen to oxidize hydrogen sulfide into sulfur dioxide, and the sulfur dioxide catalytically reacts with hydrogen sulfide to produce elemental sulfur. The existence of some compounds, such as benzene, toluene, and xylene (also referred to as BTX) in the gas stream can be detrimental to the process. For example, the presence of BTX can interfere with the catalytic conversion of sulfur dioxide into elemental sulfur.

SUMMARY

(3) This disclosure describes technologies relating to removal of BTX from acid gas in sulfur recovery units (SRUs), and in particular, adaptive control of bypass valves for minimizing moisture carryover, thereby mitigating and/or eliminating SRU furnace flameout and unit trips. Certain aspects of the subject matter described can be implemented as a system. The system includes an acid gas stream, an outlet pipe manifold, multiple activated carbon beds, a furnace, and a control system. For each activated carbon bed, the system includes an outlet flowline and a bypass flowline. The acid gas stream includes hydrogen sulfide, benzene, toluene, and xylene. Each activated carbon bed is configured to remove benzene, toluene, and xylene (BTX) from the acid gas stream in response to contacting the acid gas stream. The activated carbon beds are in a parallel flow configuration. Each outlet flowline connects the respective activated carbon bed to the outlet pipe manifold. Each outlet flowline includes an outlet valve that is configured to control flow of the acid gas stream exiting the respective activated carbon bed through the respective outlet flowline. Each bypass flowline is connected to the respective outlet flowline. Each bypass flowline has an inner diameter that is smaller in comparison to an inner diameter of the respective outlet flowline. Each bypass flowline provides an alternative flow path for the acid gas stream exiting the respective activated carbon bed around the respective outlet flowline. Each bypass flowline includes a bypass valve that is configured to control flow of the acid gas stream exiting the respective activated carbon bed through the respective bypass flowline. The furnace is in fluid

communication with the outlet pipe manifold. The furnace is configured to receive the acid gas stream from the outlet pipe manifold. The furnace includes a burner. The burner is configured to combust at least a portion of the acid gas stream in the presence of oxygen. The control system includes a temperature sensor. The temperature sensor is coupled to the furnace. The control system is communicatively coupled to each bypass valve. The temperature sensor is configured to measure a furnace temperature of the furnace. The control system is configured to adjust a percent opening of at least one of the bypass valves based on the furnace temperature measured by the temperature sensor.

(4) This, and other aspects, can include one or more of the following features. The temperature sensor can be configured to transmit the measured furnace temperature. The control system can include a controller that is communicatively coupled to the temperature sensor and to each bypass valve. The controller can include one or more processors and a non-transitory computer-readable storage medium that is coupled to the one or more processors. The storage medium can store programming instructions by execution by the one or more processors. The programming instructions can instruct the one or more processors to perform operations. The operations can include receiving the measured furnace temperature from the temperature sensor. The operations can include determining a difference between the measured furnace temperature and a specified setpoint temperature. The operations can include transmitting a close signal to at least one of the bypass valves to reduce the percent opening of the respective bypass valve in response to determining that the difference between the measured furnace temperature and the specified setpoint temperature is equal to or greater than 10 degrees Fahrenheit ($^{\circ}$ F.) differential. For each 10° F. differential in the difference between the measured furnace temperature and the specified setpoint temperature, the controller can be configured to reduce the percent opening of the respective bypass valve by 10% down to a minimum percent opening of 50%. The specified setpoint temperature can be about 1,950 $^{\circ}$ F. For each activated carbon bed, a ratio of the inner diameter of the bypass flowline to the inner diameter of the outlet flowline can be in a range of from 6:32 to 10:28. For each outlet flowline, a ratio of the inner diameter of the respective outlet flowline to an inner diameter of the outlet pipe manifold can be in a range of from 28:44 to 32:40. The system can include a preheater installed on the outlet pipe manifold upstream of the furnace. The preheater can be configured to heat the acid gas stream prior to the acid gas stream entering the furnace. The system can include a main bypass flowline connected to the outlet pipe manifold and the furnace. The main bypass flowline can provide an alternative flow path for the acid gas stream to bypass the preheater and the burner. The main bypass flowline can include a main bypass valve. The main bypass valve can be configured to control flow of the acid gas stream through the main bypass flowline.

(5) Certain aspects of the subject matter described can be implemented as a system. The system includes an outlet flowline from an activated carbon bed, a bypass flowline, a furnace, and a control system. The outlet flowline is configured to flow an acid gas stream out of the activated carbon bed after the activated carbon bed has removed benzene, toluene, and xylene from the acid gas stream. The bypass flowline is connected to the outlet flowline. The bypass flowline has an inner diameter that is smaller in comparison to an inner diameter of the outlet flowline. The bypass flowline provides an alternative flow path for the acid gas stream exiting the activated carbon bed around the outlet flowline. The bypass flowline includes a bypass valve. The bypass valve is configured to control flow of the acid gas stream exiting the activated carbon bed through the bypass flowline. The furnace is configured to receive the acid gas stream from at least one of the outlet flowline or the bypass flowline. The furnace includes a burner. The burner is configured to combust at least a portion of the acid gas stream in the presence of oxygen. The control system includes a temperature sensor and a controller. The temperature sensor is coupled to the furnace. The temperature sensor is configured to measure a furnace temperature of the furnace. The temperature sensor is configured to transmit the measured furnace temperature. The controller is

communicatively coupled to the temperature sensor and to the bypass valve. The controller is configured to receive the measured furnace temperature from the temperature sensor. The controller is configured to determine a difference between the measured furnace temperature and a specified setpoint temperature. The controller is configured to reduce a percent opening of the bypass valve in response to determining that the difference between the measured furnace temperature and the specified setpoint temperature is equal to or greater than 10° F. differential.

(6) This, and other aspects, can include one or more of the following features. For each 10° F. differential in the difference between the measured furnace temperature and the specified setpoint temperature, the controller can be configured to reduce the percent opening of the bypass valve by 10% down to a minimum percent opening of 50%. The specified setpoint temperature can be about 1,950° F. A ratio of the inner diameter of the bypass flowline to the inner diameter of the outlet flowline can be in a range of from 6:32 to 10:28. A ratio of the inner diameter of the outlet flowline to an inner diameter of the outlet pipe manifold can be in a range of from 28:44 to 32:40. The system can include a preheater installed on the outlet pipe manifold upstream of the furnace. The preheater can be configured to heat the acid gas stream prior to the acid gas stream entering the furnace. The system can include a main bypass flowline connected to the outlet pipe manifold and the furnace. The main bypass flowline can provide an alternative flow path for the acid gas stream to bypass the preheater and the burner. The main bypass flowline can include a main bypass valve. The main bypass valve can be configured to control flow of the acid gas stream through the main bypass flowline.

(7) Certain aspects of the subject matter described can be implemented as a method. An acid gas stream is flowed to an activated carbon bed. The acid gas stream includes hydrogen sulfide, benzene, toluene, and xylene. The activated carbon bed removes benzene, toluene, and xylene from the acid gas stream. The acid gas stream is flowed from the activated carbon bed to a furnace. The furnace includes a burner. The burner combusts at least a portion of the acid gas stream in the presence of oxygen. A temperature sensor coupled to the furnace measures a furnace temperature of the furnace. A controller communicatively coupled to the temperature sensor determines a difference between the measured furnace temperature and a specified setpoint temperature. The controller adjusts flow of the acid gas stream from the activated carbon bed to the furnace in response to determining that the difference between the measured furnace temperature and the specified setpoint temperature is equal to or greater than 10° F. differential.

(8) This, and other aspects, can include one or more of the following features. The activated carbon bed can include an outlet flowline and a bypass flowline that is connected to the outlet flowline. The bypass flowline can have an inner diameter that is smaller in comparison to an inner diameter of the outlet flowline. The bypass flowline can provide an alternative flow path for the acid gas stream exiting the activated carbon bed around the outlet flowline. The bypass flowline can include a bypass valve. Flowing the acid gas stream from the activated carbon bed to the furnace can include flowing at least a portion of the acid gas stream from the activated carbon bed through the bypass flowline to the furnace. Adjusting flow of the acid gas stream from the activated carbon bed to the furnace can include transmitting, by the controller, a close signal to the bypass valve to reduce a percent opening of the bypass valve in response to determining that the difference between the measured furnace temperature and the specified setpoint temperature is equal to or greater than 10° F. differential. For each 10° F. differential in the difference between the measured furnace temperature and the specified setpoint temperature, the close signal transmitted by the controller can cause the percent opening of the bypass valve to reduce by 10% down to a minimum percent opening of 50%. The specified setpoint temperature can be about 1,950° F.

(9) The details of one or more implementations of the subject matter of this disclosure are set forth in the accompanying drawings and the description. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

Description

DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic diagram of an example system including removal of BTX from acid gas in an SRU.
- (2) FIG. 2 is a flow chart of an example method for adaptive control of BTX removal from acid gas in an SRU.

DETAILED DESCRIPTION

- (3) This disclosure describes removal of BTX from acid gas in SRUs. The BTX removal is implemented by distributing the acid gas through activated carbon beds. At least two activated carbon beds are included, such that one of the beds may be put on a regeneration cycle to be reactivated while the other bed(s) are kept online. The beginning of acid gas admission through the activated carbon beds to the furnace has the greatest risk and extent of moisture carryover, which can detrimentally result in furnace flameout and unit downtime.
- (4) The subject matter described in this disclosure can be implemented in particular implementations, so as to realize one or more of the following advantages. The process is equipped with a control scheme that provides adaptive control for bypass control valves linked to the SRU reaction furnace temperature to minimize moisture carryover, thereby mitigating and/or eliminating SRU furnace flameout and unit trips. The control schemes described can mitigate and/or eliminate the potential of activating emergency shutdown by gradually admitting the acid gas through the smaller bypass control valve(s) first before opening the larger outlet valve(s) to the SRU reaction furnace to mitigate and/or eliminate the risk of sudden, high moisture content from reaching the SRU reaction furnace, which could detrimentally lead to flameout and activation of unit emergency shutdown. The systems and processes described can prevent unnecessary activation of emergency shutdown bypasses during acid gas admission to the SRU furnace and can therefore ensure safe and reliable SRU operations. The control schemes described can mitigate and/or eliminate BTX and hydrocarbon carryover, which can detrimentally deactivate and shorten the operating life of SRU catalyst(s) downstream of the SRU reaction furnace. Thus, the processes and systems described can improve sulfur recovery operations while remaining within safe environmental emissions limits.
- (5) FIG. 1 is a schematic diagram of an example system **100** including BTX removal from acid gas in an SRU. The system **100** can, for example, be an SRU. The system **100** includes an acid gas stream **101a**, multiple activated carbon beds **110**, an outlet pipe manifold **120**, a furnace **130**, and a control system **140**. Each activated carbon bed is labeled with reference number **110** followed by a letter for differentiation. In some implementations, as shown in FIG. 1, the system **100** includes two activated carbon beds **110a**, **110b**. Although shown in FIG. 1 as including two activated carbon beds (**110a**, **110b**), the system **100** can include only one activated carbon bed or more than two activated carbon beds (for example, three, four, or more than four). The activated carbon beds **110** are in a parallel flow configuration. In some implementations, the system **100** can include at least two activated carbon beds (**110a**, **110b**), such that at least one of the activated carbon beds can be reactivated while the remaining activated carbon bed(s) remain online, so that the system **100** does not need to be taken offline for reactivation of the carbon beds **110**.
- (6) Each activated carbon bed (**110a**, **110b**) is configured to remove BTX from the acid gas stream **101a** in response to contacting the acid gas stream **101a**. Each activated carbon bed (**110a**, **110b**) includes a bed of activated carbon. For each of the activated carbon beds (**110a**, **110b**), as the acid gas stream **101a** flows through, BTX adsorbs to the activated carbon. The acid gas stream **101b** exiting the activated carbon beds **110** has a reduced BTX content in comparison to the acid gas stream **101a** entering the activated carbon beds **110**. For example, the acid gas stream **101b** exiting the activated carbon beds **110** is substantially free of BTX or includes only a negligible amount of BTX. In some implementations, the system **100** includes a heater (not shown) upstream of the

activated carbon beds **110**. The heater can be configured to pre-heat the acid gas stream **101a** prior to entering the activated carbon beds **110**. Pre-heating the acid gas stream **101a** can, for example, facilitate removal of BTX by the activated carbon beds **110**.

(7) Each activated carbon bed (**110a**, **110b**) includes an outlet flowline (**111a**, **111b**, respectively) and a bypass flowline (**113a**, **113b**). Each outlet flowline (**111a**, **111b**) connects the respective activated carbon bed (**110a**, **110b**) to the outlet pipe manifold **120**. Each outlet flowline (**111a**, **111b**) includes an outlet valve (**112a**, **112b**, respectively). Each outlet valve (**112a**, **112b**) is configured to control flow of the acid gas stream **101b** exiting the respective activated carbon bed (**110a**, **110b**) through the respective outlet flowline (**111a**, **111b**). Each bypass flowline (**113a**, **113b**) is connected to the respective outlet flowline (**111a**, **111b**). Each bypass flowline (**113a**, **113b**) has an inner diameter that is smaller in comparison to an inner diameter of the respective outlet flowline (**111a**, **111b**). Each bypass flowline (**113a**, **113b**) provides an alternative flow path for the acid gas stream **101b** exiting the respective activated carbon bed (**110a**, **110b**) around the respective outlet flowline (**111a**, **111b**). Each bypass flowline (**113a**, **113b**) includes a bypass valve (**114a**, **114b**, respectively). Each bypass valve (**114a**, **114b**) is configured to control flow of the acid gas stream **101b** exiting the respective activated carbon bed (**110a**, **110b**) through the respective bypass flowline (**113a**, **113b**). In some implementations, for each activated carbon bed (**110a**, **110b**), a ratio of the inner diameter of the bypass flowline (**113a**, **113b**) to the inner diameter of the respective outlet flowline (**111a**, **111b**) is in a range of from 6:32 to 10:28. For example, for each activated carbon bed (**110a**, **110b**), the inner diameter of the bypass flowline (**113a**, **113b**) is about 8 inches, and the inner diameter of the respective outlet flowline (**111a**, **111b**) is about 30 inches (ratio of about 8:30). In some implementations, for each outlet flowline (**111a**, **111b**), a ratio of the inner diameter of the respective outlet flowline (**111a**, **111b**) to an inner diameter of the outlet pipe manifold **120** is in a range of from 28:44 to 32:40. For example, for each outlet flowline (**111a**, **111b**), the inner diameter of the respective outlet flowline (**111a**, **111b**) is about 30 inches, and the inner diameter of the outlet pipe manifold **120** is about 42 inches (ratio of about 30:42). In some implementations, the outlet pipe manifold **120** is sized to accommodate a flow rate of the acid gas stream **101b** in a range of from about 45 million standard cubic feet per day (MMSCFD) to about 70 MMSCFD.

(8) The furnace **130** is in fluid communication with the outlet pipe manifold **120**. The furnace **130** is configured to receive the acid gas stream **101b** from the outlet pipe manifold **120**. The furnace **130** includes a burner **131**. The burner **131** is configured to combust at least a portion of the acid gas stream **101b** in the presence of oxygen. Combustion of the acid gas stream **101b** within the furnace **130** oxidizes the hydrogen sulfide into sulfur dioxide. The sulfur dioxide can be catalytically converted to elemental sulfur, for example, in a downstream catalytic bed convertor.

(9) The control system **140** includes a temperature sensor **141**. The temperature sensor **141** is coupled to the furnace **130**. The temperature sensor **141** is configured to measure a furnace temperature of the furnace **130**. The control system **140** is communicatively coupled to each bypass valve (**114a**, **114b**). The control system **140** is configured to adjust a percent opening of at least one of the bypass valves (**114a**, **114b**) based on the furnace temperature measured by the temperature sensor **141**. The control system **140** can include a controller **142**. The controller **142** can be communicatively coupled to the temperature sensor **141** and each bypass valve (**114a**, **114b**). The temperature sensor **141** can be configured to transmit the measured furnace temperature, for example, to the controller **142**.

(10) The controller **142** can include one or more processors **143** and a non-transitory computer-readable storage medium (memory) **144**. **142** The processor **143** may be a microprocessor, a multi-core processor, a multithreaded processor, an ultra-low-voltage processor, an embedded processor, or a virtual processor. In some implementations, the processor **143** may be part of a system-on-a-chip (SoC) in which the processor **143** and the other components of the controller **142** are formed into a single integrated electronics package. In some implementations, the processor **143** may

include processors from Intel® Corporation of Santa Clara, California, from Advanced Micro Devices, Inc. (AMD) of Sunnyvale, California, or from ARM Holdings, LTD., Of Cambridge, England. Any number of other processors from other suppliers may also be used. Generally, the processor **143** executes instructions and manipulates data to perform the operations of the controller **142** and any algorithms, methods, functions, processes, flows, and procedures as described in this specification. The processor **143** may communicate with other components of the controller **142** over a bus. The bus may include any number of technologies, such as industry standard architecture (ISA), extended ISA (EISA), peripheral component interconnect (PCI), peripheral component interconnect extended (PCIx), PCI express (PCIe), or any number of other technologies. The bus may be a proprietary bus, for example, used in an SoC based system. Other bus technologies may be used, in addition to, or instead of, the technologies above.

(11) The memory **144** is coupled to the one or more processors **143** and stores programming instructions for execution by the one or more processors **143**. The programming instructions instruct the one or more processors **143** to perform operations. The memory **144** can hold data for the controller **142** or other components (or a combination of both) that can be connected to the network. While memory **144** is illustrated as an integral component of the controller **142**, the memory **144** can be external to the controller **142**. The memory **144** can be a transitory or non-transitory storage medium. In some implementations, such as in PLCs and other process control units, the memory **144** is integrated with a database used for long-term storage of programs and data. The memory **144** can include any number of volatile and nonvolatile memory devices, such as volatile random-access memory (RAM), static random-access memory (SRAM), flash memory, and the like. In smaller devices, such as PLCs, the memory **144** may include registers associated with the processor **143** itself.

(12) The controller **142** can receive the measured furnace temperature from the temperature sensor **141**. The controller **142** can determine a difference between the measured furnace temperature and a specified setpoint temperature. The controller **142** can transmit a close signal to at least one of the bypass valves (**114a**, **114b**) to reduce the percent opening of the respective bypass valve (**114a**, **114b**) in response to determining that the difference between the measured furnace temperature and the specified setpoint temperature is equal to or greater than 10 degrees Fahrenheit (° F.) differential. In some implementations, the specified setpoint temperature is about 1,950° F. (for example, 1,920° F., 1,930° F., 1,940° F., 1,950° F., 1,960° F., or 1,970° F.). In some implementations, for each 10° F. differential in the difference between the measured furnace temperature and the specified setpoint temperature, the controller **142** is configured to reduce the percent opening of the respective bypass valve (**114a**, **114b**) by 10% down to a minimum percent opening of 50%. For example, if the specified setpoint temperature is 1,950° F. and the controller **142** determines that the furnace temperature measured by the temperature sensor **141** is equal to or less than 1,940° F., the controller **142** transmits the close signal to at least one of the bypass valves (**114a**, **114b**) to reduce the percent opening of the respective bypass valve (**114a**, **114b**) by 10%, but the resulting percent opening of the respective bypass valve (**114a**, **114b**) will be no less than 50%. As another example, if the specified setpoint temperature is 1,950° F. and the controller **142** determines that the furnace temperature measured by the temperature sensor **141** is equal to or less than 1,930° F., the controller **142** transmits the close signal to at least one of the bypass valves (**114a**, **114b**) to reduce the percent opening of the respective bypass valve (**114a**, **114b**) by 20%, but the resulting percent opening of the respective bypass valve (**114a**, **114b**) will be no less than 50%. As another example, if the specified setpoint temperature is 1,950° F. and the controller **142** determines that the furnace temperature measured by the temperature sensor **141** is equal to or less than 1,920° F., the controller **142** transmits the close signal to at least one of the bypass valves (**114a**, **114b**) to reduce the percent opening of the respective bypass valve (**114a**, **114b**) by 30%, but the resulting percent opening of the respective bypass valve (**114a**, **114b**) will be no less than 50%. As another example, if the specified setpoint temperature is 1,950° F. and the controller **142**

determines that the furnace temperature measured by the temperature sensor **141** is equal to or less than 1,910° F., the controller **142** transmits the close signal to at least one of the bypass valves (**114a**, **114b**) to reduce the percent opening of the respective bypass valve (**114a**, **114b**) by 40%, but the resulting percent opening of the respective bypass valve (**114a**, **114b**) will be no less than 50%. As another example, if the specified setpoint temperature is 1,950° F. and the controller **142** determines that the furnace temperature measured by the temperature sensor **141** is equal to or less than 1,900° F., the controller **142** transmits the close signal to at least one of the bypass valves (**114a**, **114b**) to reduce the percent opening of the respective bypass valve (**114a**, **114b**) by 50%, but the resulting percent opening of the respective bypass valve (**114a**, **114b**) will be no less than 50%. Once the furnace temperature has returned to the specified setpoint temperature, the percent opening of the bypass valve(s) (**114a**, **114b**) can be increased (for example, back to 100% (fully open)).

(13) In some implementations, the system **100** includes a preheater **150**. The preheater **150** can be installed on the outlet pipe manifold **120** upstream of the furnace **130**. The preheater **150** can be configured to heat the acid gas stream **101b** prior to the acid gas stream **101b** entering the furnace **130**. In some implementations, the system **100** includes a main bypass flowline **121**. The main bypass flowline **121** can be connected to the outlet pipe manifold **120** and the furnace **130**. The main bypass flowline **121** can provide an alternative flow path for the acid gas stream **101b** to bypass the preheater **150** and the burner **131** of the furnace **130**. The main bypass flowline **121** can include a main bypass valve **122**. The main bypass valve **122** can be configured to control flow of the acid gas stream **101b** through the main bypass flowline **121**. In some implementations, the acid gas stream **101b** splits across the outlet pipe manifold **120** and the main bypass flowline **121**. For example, under normal operation, about $\frac{2}{3}$ of the acid gas stream **101b** flows through the main bypass flowline **121**, and the remaining portion of the acid gas stream **101b** flows through the outlet pipe manifold **120**. The percent opening of the main bypass valve **122** can be, for example, controlled by the control system **140**.

(14) FIG. 2 is a flow chart of an example method **200** for adaptive control of BTX removal from acid gas in an SRU. The system **100** can, for example, implement the method **200**. At block **202**, an acid gas stream (such as the acid gas stream **101a**) is flowed to an activated carbon bed (such as the activated carbon bed **110a**). In this scenario, the second activated carbon bed **110b** can be undergoing a reactivation process to reactivate the carbon bed. At block **204**, the activated carbon bed **110a** removes BTX from the acid gas stream **101a**. At block **206**, the acid gas stream **101b** (BTX removed) is flowed from the activated carbon bed **110a** to a furnace (such as the furnace **130**). As described previously, the furnace **130** includes the burner **131**. At block **208**, the burner **131** combusts at least a portion of the acid gas stream **101b** in the presence of oxygen. Combusting the acid gas stream **101b** at block **208** oxidizes the hydrogen sulfide to produce sulfur dioxide. The sulfur dioxide can be catalytically converted to elemental sulfur. At block **210**, a temperature sensor (such as the temperature sensor **141** coupled to the furnace **130**) measures a furnace temperature of the furnace **130**. At block **212**, a controller (such as the controller **142** communicatively coupled to the temperature sensor **141**) determines a difference between the measured furnace temperature (block **210**) and a specified setpoint temperature. At block **214**, the controller **142** adjusts flow of the acid gas stream **101b** from the activated carbon bed **110a** to the furnace **130** in response to determining that the difference between the measured furnace temperature and the specified setpoint temperature is equal to or greater than 10° F. differential.

(15) In some implementations, flowing the acid gas stream **101b** from the activated carbon bed **110a** to the furnace **130** at block **206** includes flowing at least a portion of the acid gas stream **101b** from the activated carbon bed **110a** through the bypass flowline **113a**. For example, during acid gas admission to the furnace **130** at block **206**, the bypass valve **114a** is open while the outlet valve **112a** is closed, such that the acid gas stream **101b** is routed through the bypass flowline **113a** as opposed to the outlet flowline **111a**. Because the bypass flowline **113a** has a smaller cross-sectional

flow area in comparison to the outlet flowline **111a** flow of the acid gas stream **101b** is limited, and sudden, high volume flow to the furnace **130** can be avoided and/or prevented. In some implementations, adjusting flow of the acid gas stream **101b** from the activated carbon bed **110a** to the furnace **130** at block **214** can include transmitting, by the controller **142**, a close signal to the bypass valve **114a** to reduce a percent opening (for example, reduce by 10%) in response to determining that the difference between the measured furnace temperature and the specified temperature is equal to or greater than 10° F. differential. The method **200** can be repeated/continued for a time duration that is considered sufficient for reaching normal operation of the SRU. For example, the method **200** can be repeated/continued for at least 20 minutes, and then the outlet valve **112a** can be gradually opened to increase flow of the acid gas stream **101b** to the furnace **130** for sulfur removal.

(16) While this specification contains many specific implementation details, these should not be construed as limitations on the scope of what may be claimed, but rather as descriptions of features that may be specific to particular implementations. Certain features that are described in this specification in the context of separate implementations can also be implemented, in combination, in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations, separately, or in any sub-combination. Moreover, although previously described features may be described as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can, in some cases, be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

(17) As used in this disclosure, the terms “a,” “an,” or “the” are used to include one or more than one unless the context clearly dictates otherwise. The term “or” is used to refer to a nonexclusive “or” unless otherwise indicated. The statement “at least one of A and B” has the same meaning as “A, B, or A and B.” In addition, it is to be understood that the phraseology or terminology employed in this disclosure, and not otherwise defined, is for the purpose of description only and not of limitation. Any use of section headings is intended to aid reading of the document and is not to be interpreted as limiting; information that is relevant to a section heading may occur within or outside of that particular section.

(18) As used in this disclosure, the term “about” or “approximately” can allow for a degree of variability in a value or range, for example, within 10%, within 5%, or within 1% of a stated value or of a stated limit of a range.

(19) As used in this disclosure, the term “substantially” refers to a majority of, or mostly, as in at least about 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, 99.9%, 99.99%, or at least about 99.999% or more.

(20) Values expressed in a range format should be interpreted in a flexible manner to include not only the numerical values explicitly recited as the limits of the range, but also to include all the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range is explicitly recited. For example, a range of “0.1% to about 5%” or “0.1% to 5%” should be interpreted to include about 0.1% to about 5%, as well as the individual values (for example, 1%, 2%, 3%, and 4%) and the sub-ranges (for example, 0.1% to 0.5%, 1.1% to 2.2%, 3.3% to 4.4%) within the indicated range. The statement “X to Y” has the same meaning as “about X to about Y,” unless indicated otherwise. Likewise, the statement “X, Y, or Z” has the same meaning as “about X, about Y, or about Z,” unless indicated otherwise.

(21) Particular implementations of the subject matter have been described. Other implementations, alterations, and permutations of the described implementations are within the scope of the following claims as will be apparent to those skilled in the art. While operations are depicted in the drawings or claims in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed (some operations may be considered optional), to achieve desirable

results. In certain circumstances, multitasking or parallel processing (or a combination of multitasking and parallel processing) may be advantageous and performed as deemed appropriate. (22) Moreover, the separation or integration of various system modules and components in the previously described implementations should not be understood as requiring such separation or integration in all implementations, and it should be understood that the described components and systems can generally be integrated together or packaged into multiple products. (23) Accordingly, the previously described example implementations do not define or constrain the present disclosure. Other changes, substitutions, and alterations are also possible without departing from the spirit and scope of the present disclosure.

Claims

1. A system comprising: an acid gas stream comprising hydrogen sulfide, benzene, toluene, and xylene; an outlet pipe manifold; a plurality of activated carbon beds, each activated carbon bed configured to remove benzene, toluene, and xylene from the acid gas stream in response to contacting the acid gas stream, the plurality of activated carbon beds in a parallel flow configuration; for each activated carbon bed: an outlet flowline connecting the respective activated carbon bed to the outlet pipe manifold, the outlet flowline comprising an outlet valve configured to control flow of the acid gas stream exiting the respective activated carbon bed through the outlet flowline; a bypass flowline connected to the outlet flowline, the bypass flowline having an inner diameter that is smaller in comparison to an inner diameter of the outlet flowline, the bypass flowline providing an alternative flow path for the acid gas stream exiting the respective activated carbon bed around the outlet flowline, the bypass flowline comprising a bypass valve configured to control flow of the acid gas stream exiting the respective activated carbon bed through the bypass flowline; a furnace in fluid communication with the outlet pipe manifold and configured to receive the acid gas stream from the outlet pipe manifold, the furnace comprising a burner configured to combust at least a portion of the acid gas stream in the presence of oxygen; a control system comprising a temperature sensor coupled to the furnace, the control system communicatively coupled to each bypass valve, the temperature sensor configured to measure a furnace temperature of the furnace, wherein the control system is configured to adjust a percent opening of at least one of the bypass valves based on the furnace temperature measured by the temperature sensor.
2. The system of claim 1, wherein the temperature sensor is configured to transmit the measured furnace temperature, and the control system comprises a controller communicatively coupled to the temperature sensor and to each bypass valve, the controller comprising: one or more processors; and a non-transitory computer-readable storage medium coupled to the one or more processors and storing programming instructions for execution by the one or more processors, the programming instructions instructing the one or more processors to perform operations comprising: receiving the measured furnace temperature from the temperature sensor; determining a difference between the measured furnace temperature and a specified setpoint temperature; and transmitting a close signal to at least one of the bypass valves to reduce the percent opening of the respective bypass valve in response to determining that the difference between the measured furnace temperature and the specified setpoint temperature is equal to or greater than 10 degrees Fahrenheit ($^{\circ}$ F.) differential.
3. The system of claim 2, wherein for each 10° F. differential in the difference between the measured furnace temperature and the specified setpoint temperature, the controller is configured to reduce the percent opening of the respective bypass valve by 10% down to a minimum percent opening of 50%.
4. The system of claim 3, wherein the specified setpoint temperature is about $1,950^{\circ}$ F.
5. The system of claim 4, wherein for each activated carbon bed, a ratio of the inner diameter of the bypass flowline to the inner diameter of the outlet flowline is in a range of from 6:32 to 10:28.
6. The system of claim 5, wherein for each outlet flowline, a ratio of the inner diameter of the

respective outlet flowline to an inner diameter of the outlet pipe manifold is in a range of from 28:44 to 32:40.

7. The system of claim 6, comprising a preheater installed on the outlet pipe manifold upstream of the furnace, the preheater configured to heat the acid gas stream prior to the acid gas stream entering the furnace.

8. The system of claim 7, comprising a main bypass flowline connected to the outlet pipe manifold and the furnace, the main bypass flowline providing an alternative flow path for the acid gas stream to bypass the preheater and the burner, the main bypass flowline comprising a main bypass valve configured to control flow of the acid gas stream through the main bypass flowline.

9. A system comprising: an outlet flowline from an activated carbon bed, the outlet flowline configured to flow an acid gas stream out of the activated carbon bed after the activated carbon bed has removed benzene, toluene, and xylene from the acid gas stream; a bypass flowline connected to the outlet flowline, the bypass flowline having an inner diameter that is smaller in comparison to an inner diameter of the outlet flowline, the bypass flowline providing an alternative flow path for the acid gas stream exiting the activated carbon bed around the outlet flowline, the bypass flowline comprising a bypass valve configured to control flow of the acid gas stream exiting the activated carbon bed through the bypass flowline; a furnace configured to receive the acid gas stream from at least one of the outlet flowline or the bypass flowline, the furnace comprising a burner configured to combust at least a portion of the acid gas stream in the presence of oxygen; a control system comprising: a temperature sensor coupled to the furnace, the temperature sensor configured to measure a furnace temperature of the furnace and transmit the measured furnace temperature; and a controller communicatively coupled to the temperature sensor and to the bypass valve, the controller configured to: receive the measured furnace temperature from the temperature sensor; determine a difference between the measured furnace temperature and a specified setpoint temperature; and reduce a percent opening of the bypass valve in response to determining that the difference between the measured furnace temperature and the specified setpoint temperature is equal to or greater than 10 degrees Fahrenheit ($^{\circ}$ F.) differential.

10. The system of claim 9, wherein for each 10° F. differential in the difference between the measured furnace temperature and the specified setpoint temperature, the controller is configured to reduce the percent opening of the bypass valve by 10% down to a minimum percent opening of 50%.

11. The system of claim 10, wherein the specified setpoint temperature is about $1,950^{\circ}$ F.

12. The system of claim 11, wherein a ratio of the inner diameter of the bypass flowline to the inner diameter of the outlet flowline is in a range of from 6:32 to 10:28.

13. The system of claim 12, wherein a ratio of the inner diameter of the outlet flowline to an inner diameter of the outlet pipe manifold is in a range of from 28:44 to 32:40.

14. The system of claim 13, comprising a preheater installed on the outlet pipe manifold upstream of the furnace, the preheater configured to heat the acid gas stream prior to the acid gas stream entering the furnace.

15. The system of claim 14, comprising a main bypass flowline connected to the outlet pipe manifold and the furnace, the main bypass flowline providing an alternative flow path for the acid gas stream to bypass the preheater and the burner, the main bypass flowline comprising a main bypass valve configured to control flow of the acid gas stream through the main bypass flowline.

16. A method comprising: flowing an acid gas stream to an activated carbon bed, the acid gas stream comprising hydrogen sulfide, benzene, toluene, and xylene; removing, by the activated carbon bed, benzene, toluene, and xylene from the acid gas stream; flowing the acid gas stream from the activated carbon bed to a furnace comprising a burner; combusting, by the burner, at least a portion of the acid gas stream in the presence of oxygen; measuring, by a temperature sensor coupled to the furnace, a furnace temperature of the furnace; determining, by a controller communicatively coupled to the temperature sensor, a difference between the measured furnace

temperature and a specified setpoint temperature; and adjusting, by the controller, flow of the acid gas stream from the activated carbon bed to the furnace in response to determining that the difference between the measured furnace temperature and the specified setpoint temperature is equal to or greater than 10 degrees Fahrenheit ($^{\circ}$ F.) differential.

17. The method of claim 16, wherein: the activated carbon bed comprises an outlet flowline and a bypass flowline connected to the outlet flowline; the bypass flowline has an inner diameter that is smaller in comparison to an inner diameter of the outlet flowline; the bypass flowline provides an alternative flow path for the acid gas stream exiting the activated carbon bed around the outlet flowline; the bypass flowline comprises a bypass valve; and flowing the acid gas stream from the activated carbon bed to the furnace comprises flowing at least a portion of the acid gas stream from the activated carbon bed through the bypass flowline to the furnace.

18. The method of claim 17, wherein adjusting flow of the acid gas stream from the activated carbon bed to the furnace comprises transmitting, by the controller, a close signal to the bypass valve to reduce a percent opening of the bypass valve in response to determining that the difference between the measured furnace temperature and the specified setpoint temperature is equal to or greater than 10° F. differential.

19. The method of claim 18, wherein for each 10° F. differential in the difference between the measured furnace temperature and the specified setpoint temperature, the close signal transmitted by the controller causes the percent opening of the bypass valve to reduce by 10% down to a minimum percent opening of 50%.

20. The method of claim 19, wherein the specified setpoint temperature is about $1,950^{\circ}$ F.
